Jackson Young

CONTACT INFORMATION

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Affiliation: School of Mathematical and Statistical Sciences, Arizona State University

EDUCATION

Ph.D. in Mathematics May 2028 (in progress)
Arizona State University Tempe, AZ

- Doctoral advisor: Dr. Julien Paupert
- GPA: 4.00

M.A. in Mathematics December 2025
Arizona State University Tempe, AZ

B.S. in Mathematics May 2023
Rensselaer Polytechnic Institute Troy, NY

- Summa Cum Laude, Dean's List. GPA: 3.94
- Minor in Computer Science

RESEARCH PAPERS

My research interests include low-dimensional geometry and topology, real and complex hyperbolic geometry, arithmetic lattices and their representations, and the structure theory of semisimple Lie groups.

Presentations for sisters of one-cusped Picard modular groups October 2026
Submitted to arXiv. Preprint.

TALKS GIVEN

The geometry of one-cusped Picard modular groups November 2026
AMS Fall Western Sectional Meeting, Arizona State University Tempe, AZ

- Upcoming

Presentations for sisters of one-cusped Picard modular groups October 2026
Geometry & Topology Seminar, Arizona State University Tempe, AZ

- Invited speaker

Presentations for the sisters of Euclidean Picard modular groups May 2026
GTA Graduate Conference, Temple University Philadelphia, PA

TEACHING EXPERIENCE

Arizona State University Fall 2023–Current
Teaching Assistant Tempe, AZ
Teaching instructor, grader, and holding office hours for a wide variety of classes, including all levels of calculus, linear algebra, real analysis, and graduate-level topology

- **Teaching Assistant** Fall 2026
  MAT 501: Geometry and Topology of Manifolds I
- **Recitation Instructor** Spring 2026
  MAT 272: Calculus with Analytic Geometry III
- **Teaching Assistant** Fall 2025
  MAT 501: Geometry and Topology of Manifolds I
- **Instructor** Summer 2025
  MAT 265: Calculus for Engineers I
- **Recitation Instructor** Spring 2025
  MAT 272: Calculus with Analytic Geometry III

- **Teaching Assistant** Fall 2024
  MAT 501: Geometry and Topology of Manifolds I
- **Instructor** Summer 2024
  MAT 265: Calculus for Engineers I
- **Teaching Assistant** Spring 2024
  MAT 343: Applied Linear Algebra
- **Recitation Instructor** Fall 2023
  MAT 271: Calculus with Analytic Geometry II

Mathnasium Summer 2023
Instructor Phoenix, AZ

- Building foundational mathematical skills in students from early elementary up to high school level calculus concepts. In charge of instructing up to four students at a time, as well as assisting with cleaning, closing, and various bookkeeping tasks

Freelance Tutoring Fall 2021–Spring 2023
Math Tutor Virtual

- Mathematics tutoring for middle and high school level students

Rensselaer I-PERSIST Program Fall 2020/Fall 2022
Mentor Troy, NY

- Host weekly mentoring sessions for first year undergraduate students, structured similarly to a recitation. Goals include helping students with college-level calculus as well as adjust to university life
- In 2020, helped redesign and adapt the freshman mentoring program to a virtual format

SELECT SUMMER SCHOOLS AND CONFERENCES

Pathways Workshop, Topological and Geometric Structures in Low Dimensions January 2026
Simons–Laufer Mathematical Sciences Institute Berkeley, CA
Funded Participant

Introductory Workshop, Topological and Geometric Structures in Low Dimensions January 2026
Simons–Laufer Mathematical Sciences Institute Berkeley, CA
Funded Participant

Hyperbolic Manifolds in Dimensions Four (And More) September 2025
Centre de Recherches Mathématiques Montreal, Quebec, Canada
Funded Participant

Topological and Geometric Structures in Low Dimensions (Summer School) July 2025
Simons–Laufer Mathematical Sciences Institute Berkeley, CA
Funded Participant

GTA Graduate Conference May 2025
Temple University Philadelphia, PA
Funded Participant

AMS Fall Western Sectional Meeting October 2024
University of California, Riverside Riverside, CA

AWARDS

SoMSS Travel Award January 2026
Arizona State University

SoMSS Travel Award August 2025
Arizona State University

LEADERSHIP

American Mathematical Society Fall 2025–Current
President, ASU Chapter

- Lead the initiative to restart the AMS graduate student chapter at Arizona State University

- Recruit members, coordinate marketing, allocate funding, draft a new consitution, organize events

Pi Mu Epislon Spring 2021–Spring 2023
Membership Coordinator, RPI Chapter

- Assist in recruitment programs and community outreach
- Formally induct all members to the national chapter

RPI Ballroom Dance Club and Team Fall 2019–Spring 2023
Captain, Competition Coordinator

- Coordinate between the university, team, and coach on team-related budgeting, marketing, private lessons, and competition registration. Contact potential judges and various officials, competing teams, venues; order materials

RPI Bridge Scholars Program Summer 2022
Mentor

- Accilmate incoming freshman, many of whome are first-generation students, to the academic and social atmosphere of RPI through teaching daily classes and running seminars

RPI Student Orientation Summer 2020
Advising Officer

- Adapt the student orientation program to the virtual format and engage incoming freshman in activities designed to meet other students and learn about the school

SKILLS

Programing Languages

- Python, LaTeX/Tikz, Mathematica, HTML/CSS, Javascript, MATLAB, C++

Professional Skills

- Management, leadership, teamwork, communication, adaptability, computer skills